

# My-N-Print: A globally harmonized, top-down nitrogen footprint calculator for food and energy choices across 181 countries

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## KEY CLAIM

*A globally accessible, web-based nitrogen footprint calculator can catalyze the same shift in nitrogen awareness that carbon tools triggered for climate*

## BRIEF JUSTIFICATION, REMAINING CHALLENGES

Carbon footprint tools have demonstrated measurable effects on consumer behavior, predicting sustainable actions (Tripathi & Trigunait, 2024), triggering behavioral spillovers (Penz et al., 2019), and steering purchasing decisions (Liu et al., 2024). My-N-Print extends this logic to nitrogen, a pollutant responsible for smog, acid rain, biodiversity loss, coastal dead zones, and climate change (Galloway et al., 2003), yet largely invisible to the public.

Key evidence from My-N-Print:

- Global nitrogen footprints range from 9.9 kg N/capita/year in Burundi to 87.2 kg N/capita/year in Saint Kitts and Nevis, with a global average of 27.2 kg N/capita/year – variation comparable to carbon.
- Food production dominates nitrogen footprints (especially beef, milk, and pork), while energy dominates carbon footprints – underlining why a nitrogen-specific tool is needed (Leach et al., 2017).
- Adopting the EAT-Lancet Planetary Health Diet globally would cut food-related nitrogen losses by ~30.5 Tg of N per year (an 18.4% decrease), demonstrating actionable mitigation potential directly linked to individual choices.
- My-N-Print covers 181 countries with a harmonized, globally consistent methodology — the first tool of its kind, building on 15+ country-specific N-Print calculators developed since 2012.

Even though My-N-Print quantifies losses to the environment, it currently lacks the means to determine the impact of these losses on the environment. Unlike carbon, nitrogen emissions can have very different impacts based on the nutrient sensitivity of the surrounding environment. Only nutrient loads that exceed safe critical thresholds are actually harmful. The main remaining challenge is to develop a method to estimate personal footprints quantified in N exceedance-caused rather than N losses.

## POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

My-N-Print sits at the consumer end of the nitrogen chain and gains meaning when connected to the farm-level, national, and regulatory perspectives represented across this session.

- With Vos (Danish N footprint) – The Danish case demonstrates how national policy can shape individual nitrogen footprints through regulation. My-N-Print can complement this by quantifying what those regulatory changes mean at the consumer level across countries, and by making the consumer's share of responsibility visible.
- With Serra (reimagining national NUE) – The input data underlying a personal nitrogen footprint (food supply, virtual nitrogen factors, production losses by commodity and country) substantially overlap with what national NUE calculations require. The key difference is normative: the footprint spans the full chain from production through consumption and assigns individual agency, while NUE focuses on the production stage as an efficiency indicator (economic as much as environmental). A reframed NUE and My-N-Print together could offer a coherent dual narrative for policymakers.
- With Oita (watershed- and SWAT-derived VNFs) - Oita's bottom-up approach directly solves the main remaining challenge described in this position paper. Oita's localized loss pathways provide the metrics that, when combined with ecological critical thresholds, can be used to translate personal footprints from abstract N losses into true, site-specific N exceedances.

## QUESTIONS TO SEED THE DISCUSSION

1. Carbon footprint calculators succeeded in part because carbon maps cleanly onto a single global metric (CO<sub>2</sub>-equivalent). Nitrogen's context-dependence means the same loss can have very different local impacts. How should we communicate a 'personal nitrogen footprint' in a way that is both scientifically credible and actionable for non-expert audiences?
2. Given that nitrogen footprints (full chain) and NUE (production stage) share much of their underlying data but differ in scope and normative framing, what would an integrated reporting framework look like — one that can serve both farm advisors focused on efficiency and policymakers focused on environmental targets?

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